

Submission Instructions:

1. Write your Full Name, Student-ID, Theory Section on top of the first page
2. Your answer should be **handwritten**
3. Submit online here: <https://forms.gle/h84ifUQaUaAv4bWZ8>
4. **NO LATE SUBMISSION WILL BE ACCEPTED.**

Q1. [2+2 Marks] Refer to the table below to answer the following questions:

Movies				Actor		
MovieID	MovieName	MovieRating	Genre	ActorID	Actor	workedOn
M02	Mission Impossible	3.5	Action	TC1	Tom Cruise	M02
M12	It	3.1	Horror	FW3	Finn Wolfhard	M12
M14	Rush Hour	4.5	Comedy	JC5	Jackie Chan	M14

- a) Insert <"M12", null, 'X', "Thriller"> into the Movies table.

Identify all constraints which will be violated by this operation and explain how you will solve it.

Answer: Key constraint is violated when we are trying to insert M12 into the movies table since M12 already exists in the table and we know every value for a key must be unique. To solve it use a unique Movie ID for instance "M13". Domain constraint is violated when we are trying to insert 'X' into the moving rating column and the datatype for X is string however from the movie rating column we can see that the domain includes the datatype double so in order to solve it the value inserted must be from the defined domain, double of the attribute moving rating.

- b) Insert <null, "Chris Evans", "M13"> into the Actor table.

Identify all constraints which will be violated by this operation and explain how you will solve it.

Answer: Entity Integrity constraint is violated when we are trying to insert null in the ActorID column which is the primary key attribute in this table and we know that the primary key attributes of each relation R cannot have null values in any tuple. In order to solve it, provide a valid unique ActorID, for instance "CE4". Referential Integrity constraint is being violated when we are trying to insert "M13" into the workedOn column because the M13 value doesn't exist in the column MovieID in Movies table which this column is being referenced to. We know that the value in the foreign key column of the referencing table needs to be an existing value in the primary key column (or columns) of the referenced table. In order to solve it, Either change the workedOn to an existing MovieID or first insert movie "M13" into the movies table before inserting it into the actor table.

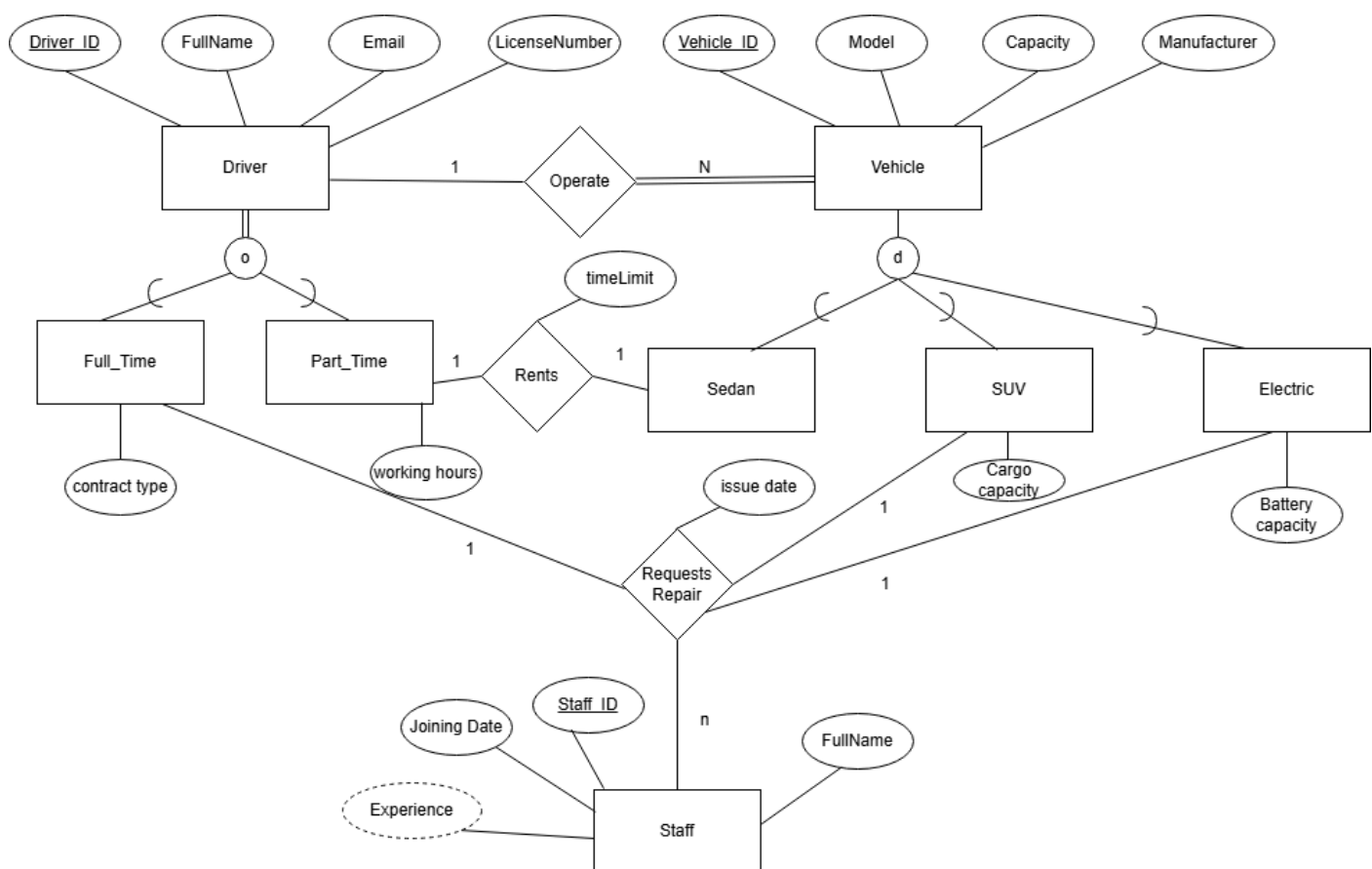
Q2. [8 Marks] You are tasked with designing a Driver Management System. The system must meet the following requirements:

- Drivers will have a unique DriverID, full name, email, and license number.
- Each vehicle has a unique VehicleID, model, capacity, and manufacturer.
- Each driver can operate many vehicles, but each vehicle can only be operated by one driver at a time.

- There are only 2 types of drivers: Full-time and Part-time. A driver can be both at the same time. Full-time drivers have a contract_type, while part-time drivers have working_hours.
- Some Vehicles are categorized by their type, such as Sedan, SUV or Electric. A vehicle can belong to only one type. Electric vehicles have BatteryCapacity, SUVs have cargoCapacity.
- A Part-time Driver can rent one Sedan at a time. When a Sedan is rented, the timeLimit is stored.
- A Full-time Driver can request repair from Staff on an SUV or an Electric. The issueDate is recorded when a repair request is made.
- A Staff will have StaffID, FullName, JoiningDate. The experience of a Staff can be calculated from JoiningDate.

Now, draw an **EER** (Extended Entity Relationship) diagram of this system. Write down all assumptions (if any).

Answer:



Q3. [8 Marks] Construct a relational Schema by mapping the following EER diagram (Figure Q3). The following conditions must be fulfilled when constructing the schema:

- For mapping the 1:N or 1:1 relationships, you **MUST** use the foreign key approach.
- For mapping the superclasses/subclasses, **select** the applicable option between 8B or 8C and **explain** briefly why the unselected option is not applicable for the given EER.

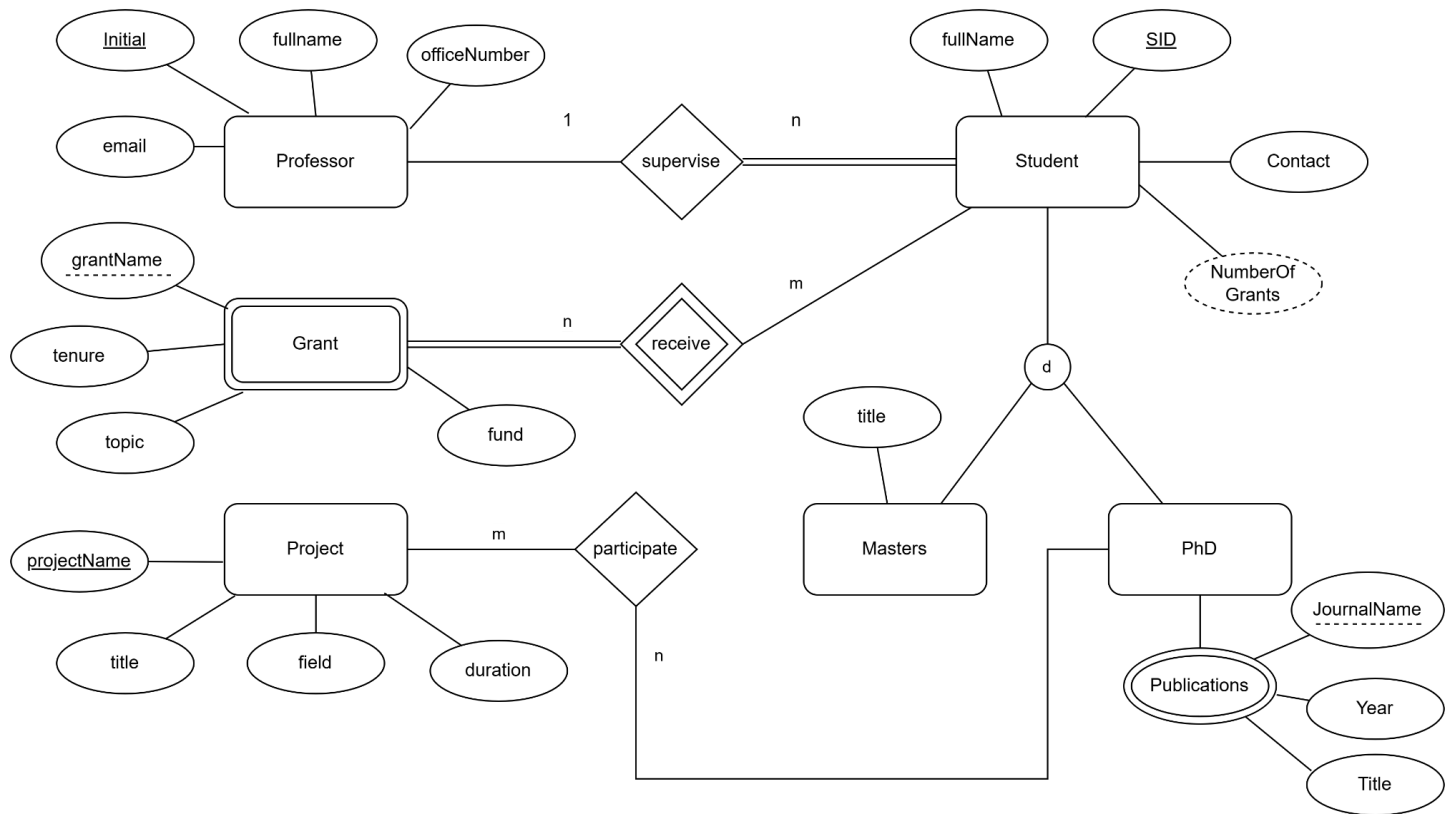


Figure: Q3

Answer:

8C Applicable. Using 8C.

8B only for total (double line in generalization).

